DO NOW — RECALL

Lesson 082 · Quicksort: Partition Around a Pivot

Name: _____ Class: _____ Date: _____

1. What is merge sort's combine step?

2. What is a pivot in quicksort?

3. Is a partitioned array necessarily fully sorted?

4. Swap values at indexes0 and2 in [4,1,7]. Give the resulting array.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Merge two individually sorted runs.

2. Model answer

Selected value used to partition an array.

3. Model answer

No; subarrays still need sorting.

4. Model answer

[7,1,4].

By the end, you can…

• Partitioning reorders values around a chosen pivot so values to its left are no larger and those right are larger, for this partition rule.

• The pivot's final index is fixed after partition, but neither side is necessarily fully sorted; quicksort continues with the two subarrays.

• Average and best-case time O(n log n), worst $O(n^2)$ with poor pivot choices; recursion consumes stack space depending on partition balance.

Vocabulary — know these terms

quicksort — Recursive partition-based sorting.

pivot — Value partitioning subarray.

partition — Place smaller/larger values around pivot.

worst case — Maximum growth for adverse input.

NEW LEARNING

Read, annotate and explain

Observe

Partitioning reorders values around a chosen pivot so values to its left are no larger and those right are larger, for this partition rule.

Apply

The pivot's final index is fixed after partition, but neither side is necessarily fully sorted; quicksort continues with the two subarrays.

Evaluate

Average and best-case time O(n log n), worst $O(n^2)$ with poor pivot choices; recursion consumes stack space depending on partition balance.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Partition [6,2,5,1,3] using last pivot3; give partition state.

Method and result

Lomuto partition moves 2 and 1 left, pivot3 to index2 → [2,1,3,6,5].

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Trace Lomuto pivot-last partition of [5,2,7,1,4] and give final pivot index. [6]

Question 2

After quicksort partition gives [2,1,4,5,7] with pivot4 at index2, state required recursive subranges and eventual sorted result. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Explain how pivot choice impacts quicksort worst-case time and recursive stack depth. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Once a pivot reaches its correct final index, both partitions are also completely sorted.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.